

Lecture 2:

Overview of Data Science & Big Data

CMSC/DATA 320: Introduction to Data Science

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Announcements

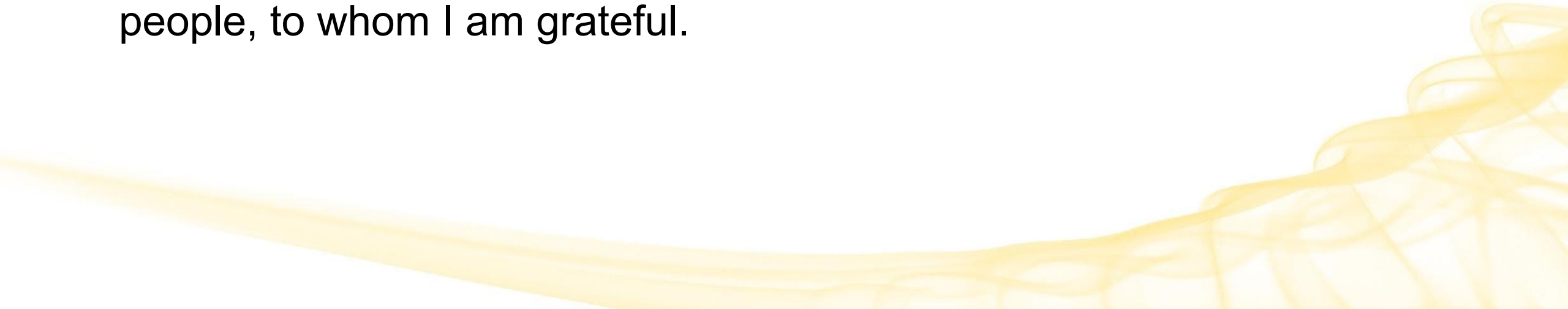
- PA1, WA1 are out, due Feb 8

Thank you

A special thank you to Tammy Perrin and Dr. Fardina Alam whose content has been used for this course.

Some slides have been handed down and refined across multiple instructors.

I've added my own flavor, and in some cases entire sections, but many of the materials have been the culmination of effort across numerous people, to whom I am grateful.



Overview of Data Science

What is Data Science?

Data Science is...

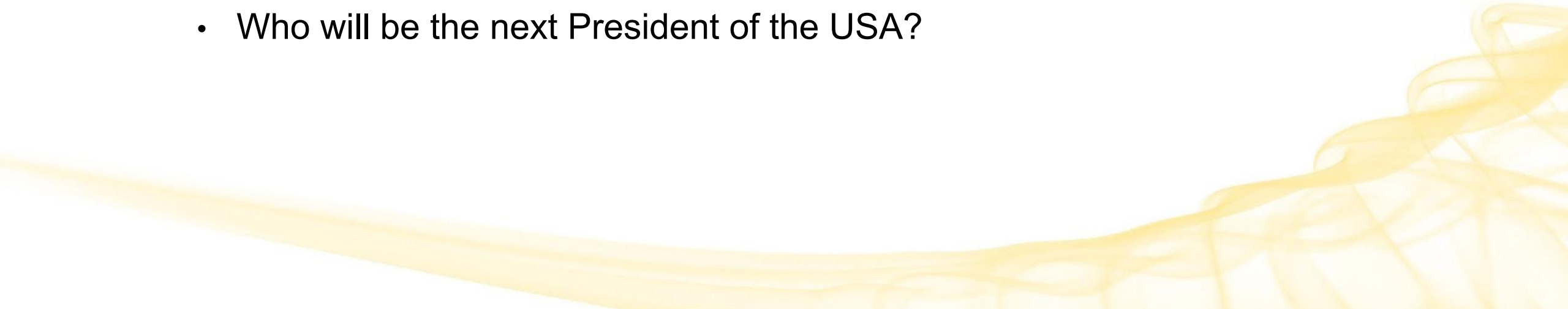
“...the application of **computational** and **statistical** techniques to address or gain [managerial or scientific] insight into some problem in the **real world**.”

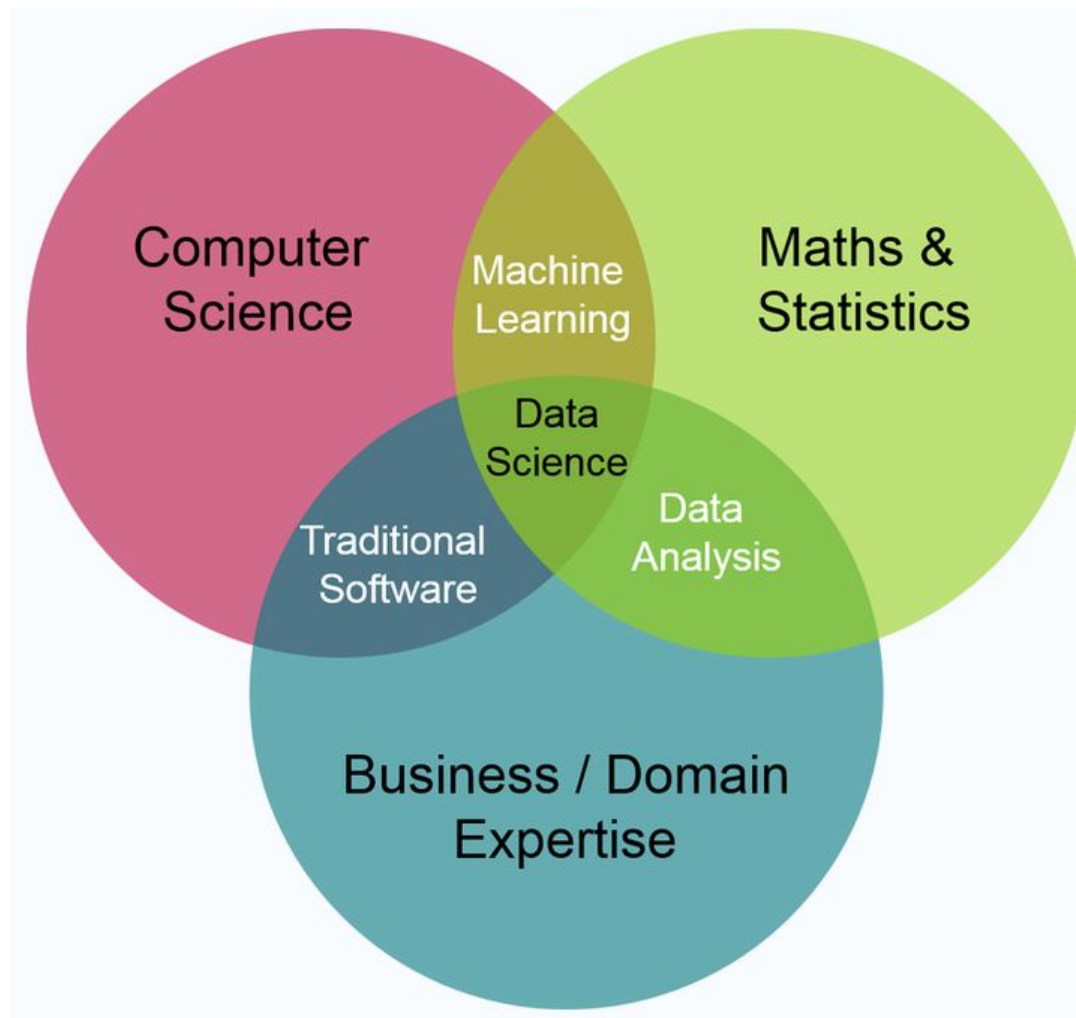
–Zico Kolter (*Machine Learning Professor at CMU and Chief Scientist at Bosch*)



Data Science is all about using data + tools to **solve problems**.

Examples:

- **Decision making**
 - Which email is spam and which is not?
 - **Product recommendations**
 - Which movie to watch?
 - **Predicting the outcome**
 - Who will be the next President of the USA?
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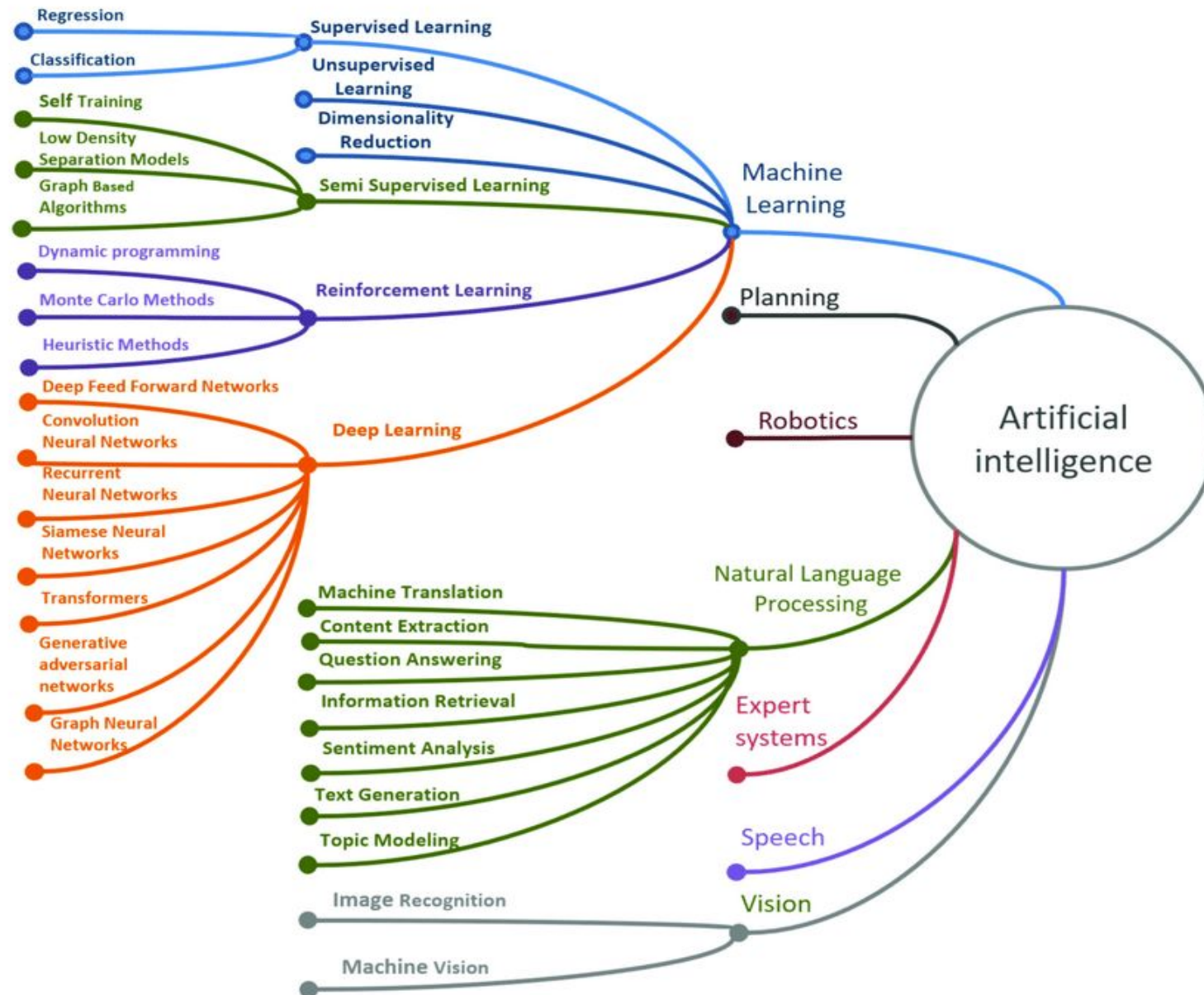


Slightly modified version of the famous Drew Conway Venn diagram
(Drew Conway is founder and CEO of Alluvium, an analytics company)

Core skills in Data Science

- Statistics, computer science, mathematics
 - Data cleaning and formatting
 - Data visualization
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What is Machine Learning?



Data Science vs Data Mining

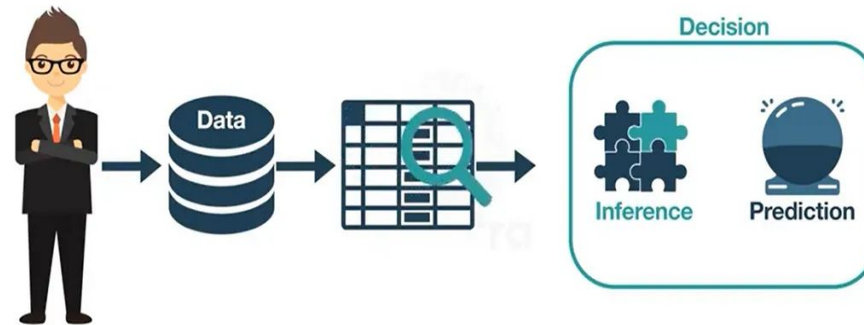
- Data mining
 - The probing of available datasets in order to identify patterns and anomalies.
- Machine learning
 - Algorithms and statistical models that enable systems to learn and improve their performance on a specific task or problem without being explicitly programmed
- Data Science
 - Uses data mining and machine learning techniques (among others!)

Data Mining is a **subset** of Data Science
It is a technique we leverage in the DS Process

This Course

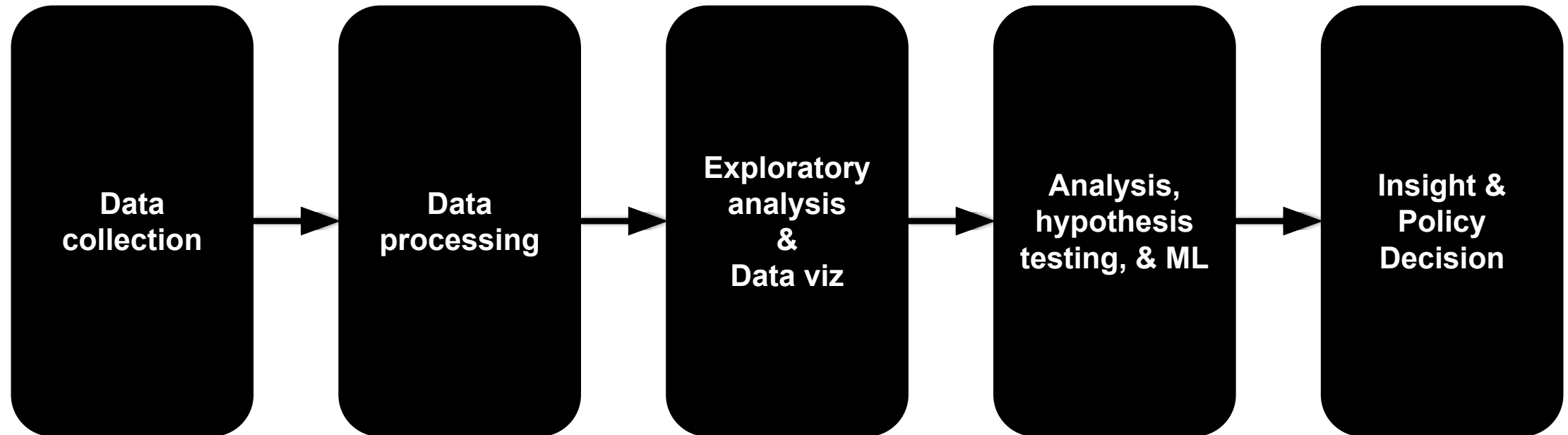
In this course, we will learn some aspects of taking **raw data** and turning it into **insights** about the world or **predictions** about the future.

We will also learn about **telling a story** to convey these insights and predictions in a useful and compelling way.



The Data Science life cycle

The **Data Science Life Cycle**, or process



1. Collect Data

Data collection is a systematic approach to gather **relevant information** from a **variety of sources**.

- external sources
- existing internal datastores and streams
- tools and processes created by you



Data Collection: primary vs secondary

Primary

Situation:

Unique problem and no related research/data collection done for it yet.

Solution:

Collect the data.

- Example: Avg time employees spend on lunch breaks across companies
- Problem: no public data available
- Solution: collect the data
- Different methods: surveys, interviews with employees, etc
- These methods are time-consuming!

Secondary

Situation:

Some problem and the data is readily available or already collected by someone else.

Solution:

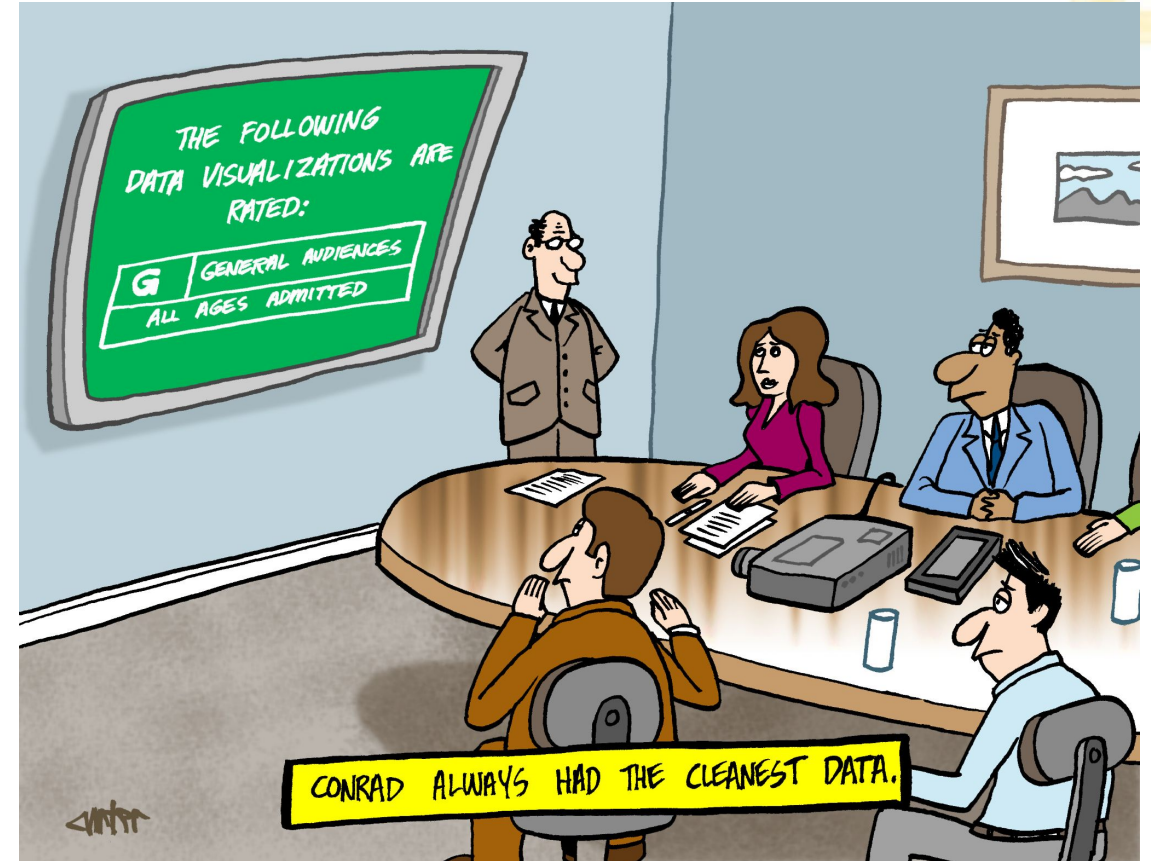
Use the data → Secondary data collection.

- Different Methods: Scrape the internet, news articles, government census, magazines and so on.
- These methods are less time consuming than the primary method.

2. Data Processing

Clean, scrub, and sanity-check data to ensure the **data quality**

- Poor data quality may lead to **unexpected results** or **misleading information**
 - Deal with duplicates
 - Formatting
 - Weird outliers
 - Mistakes
 - Transformations needed
 - Sampling

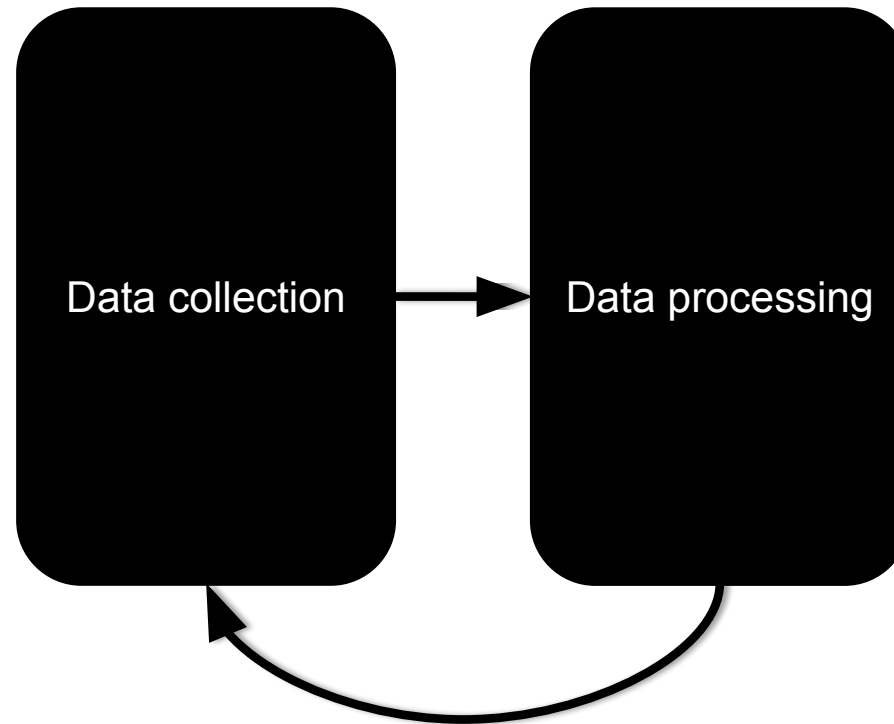


A simple Data Processing example

You collect data about students' test scores.

- **Problem:** Some scores left blank, others wrote "N/A" (Missing Values).
- **Solution:** Process the Data
 - Expensive: go back to the data source to find the correct values for the missing scores
 - Cheaper:
 - Replace "N/A" entries with a neutral value, maybe 0 or something else
 - Could also be reasonable: drop the data with missing values

The Data Life Cycle



3. Explore & Visualize Data



Explore data

- Figure out what you have!
 - What is there?
 - Any interesting correlations/relationships?
 - Do you have everything you need?
- Extract useful insights
 - Understand patterns and relationships
 - Set the stage for effective model building and decision-making
- Analyze and build familiarity with the data
 - Understand the shape and spread of attributes

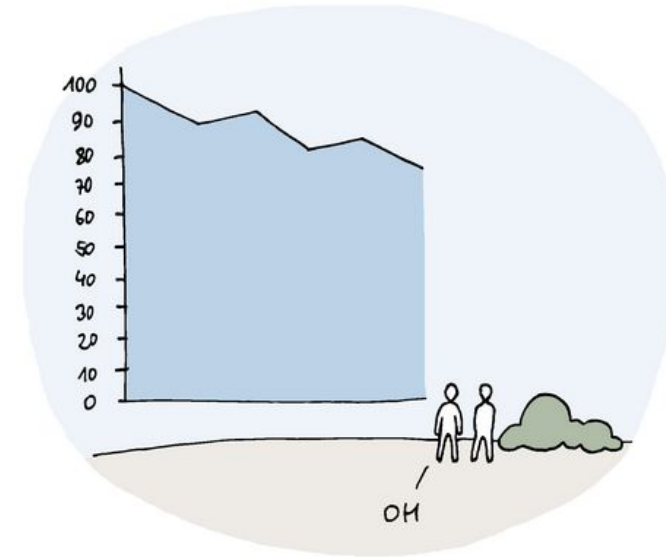
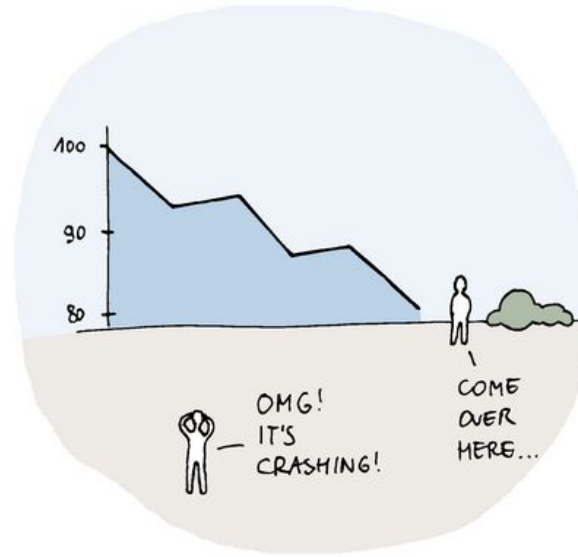


*Skipping data exploration may lead to **inaccurate models** and **insignificant variables** in your models!*

Visualize the Data

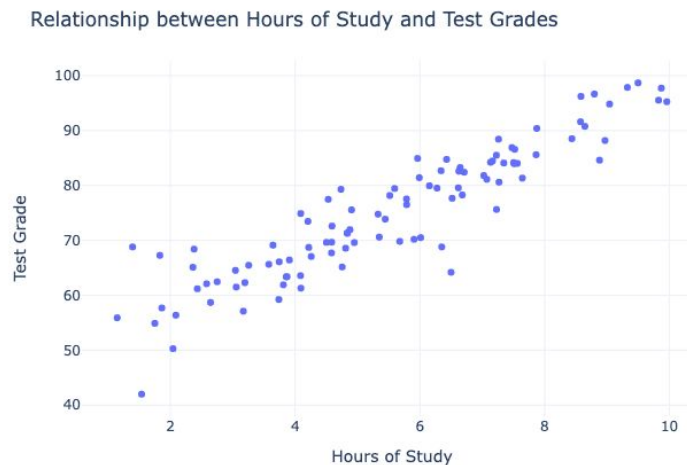
Data visualization is the process of creating graphical representations of information

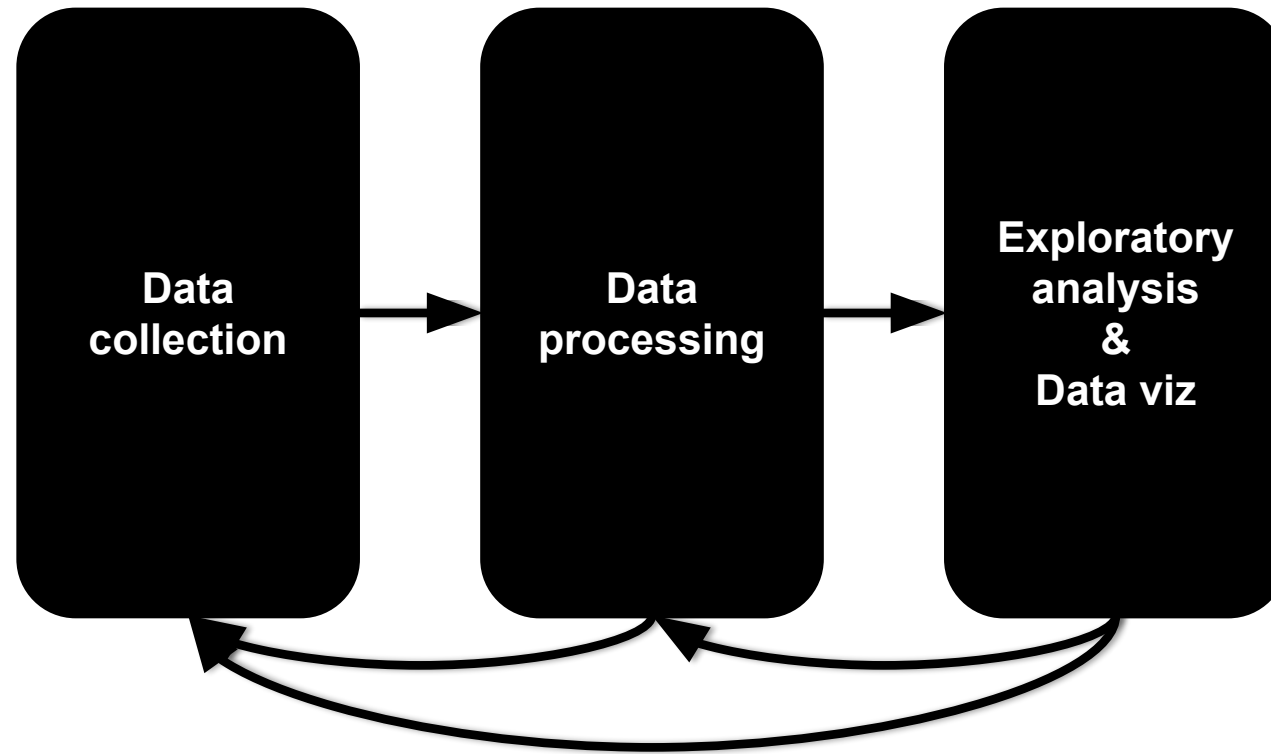
- Visualizations can help us see patterns and trends that aren't always obvious
- Can also help us communicate to others what we are seeing
 - Sometimes this results in more funding!
- Visualize things like
 - The relationship between attributes
 - High-dimensional data sets
- Be careful! Visualizations can also be misleading if not done well



Exploring our student test scores example

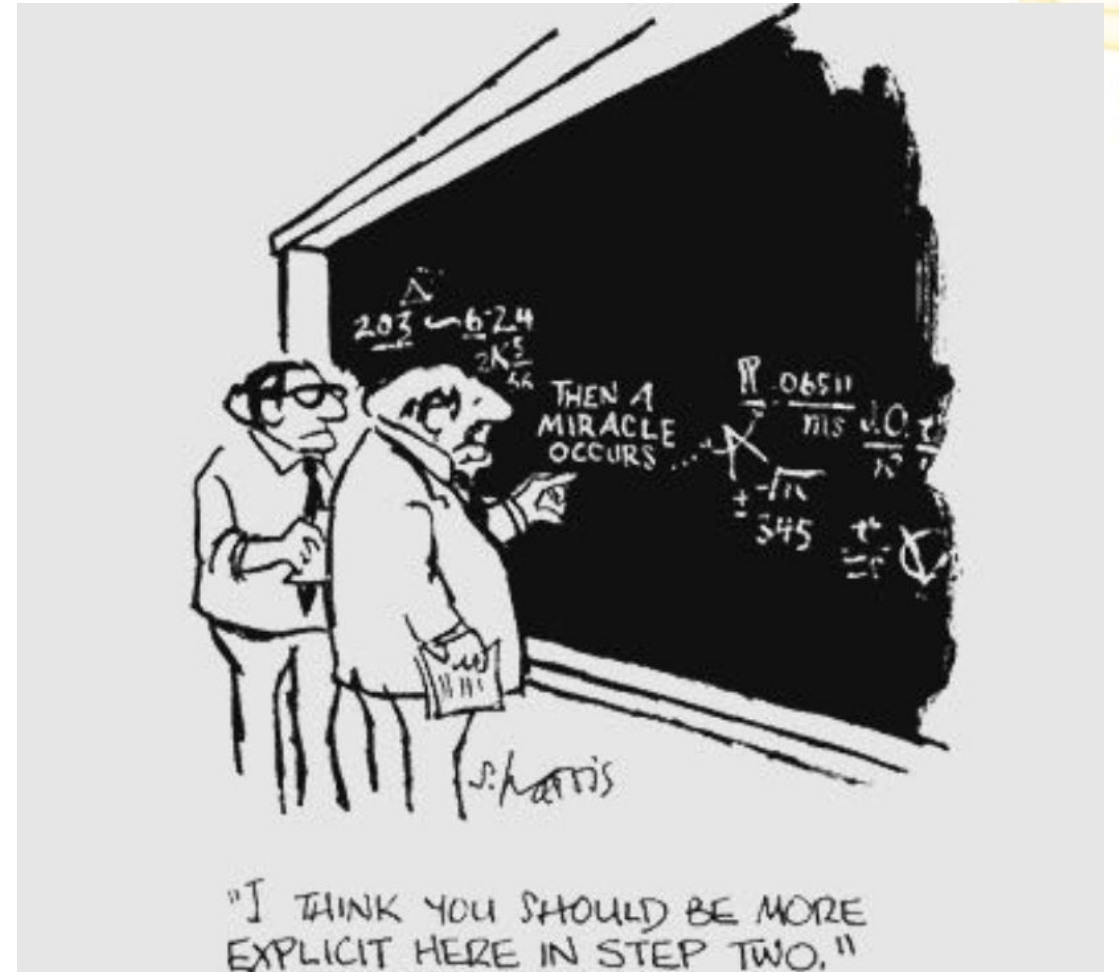
- **Exploratory Analysis:** Explore connections between study hours and scores. Calculate the average score, find the range of scores, and notice that some students scored exceptionally well.
- **Data Visualization:** Visually show the relationship between study hours and average scores for different groups of students in order to reveal that students who study more tend to have higher average scores.





4. Build a Model

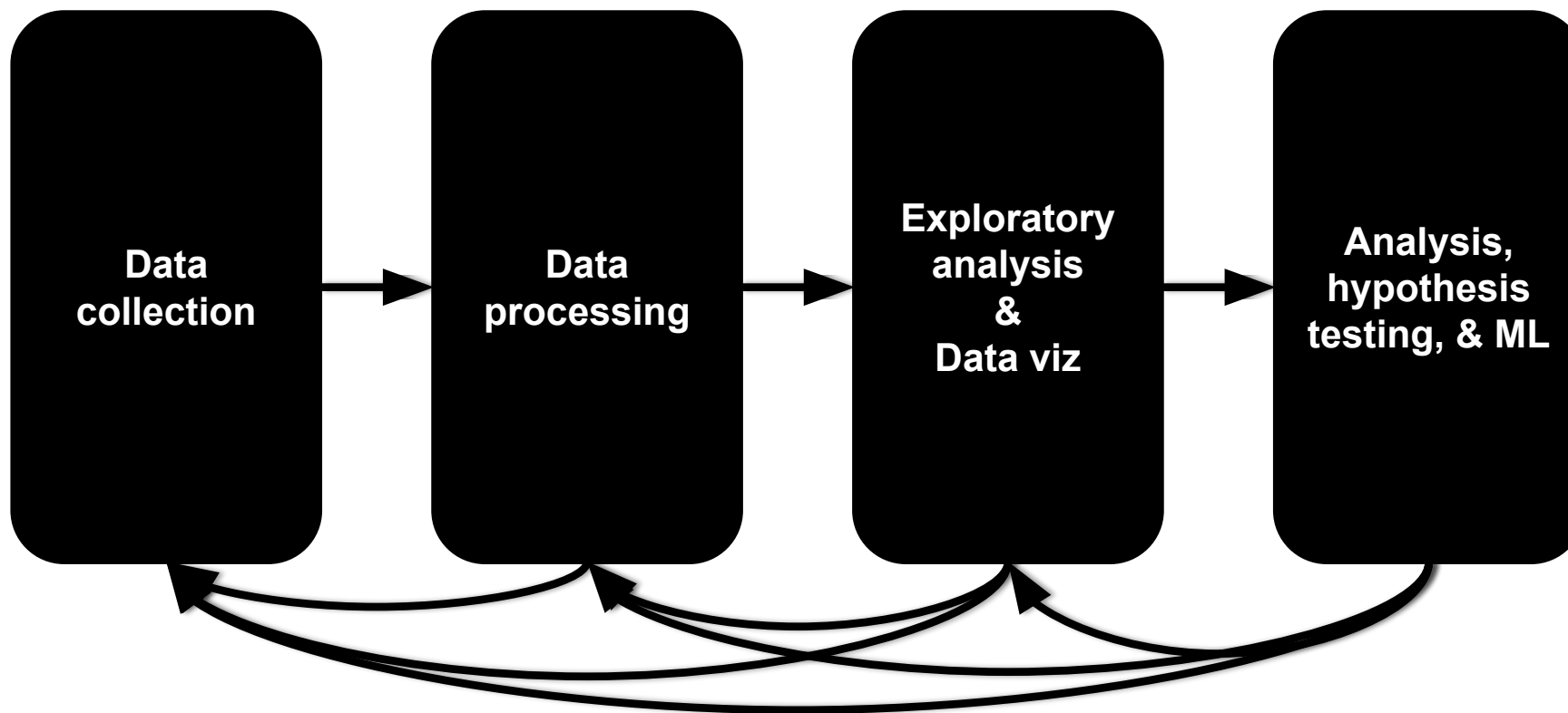
A model is a mathematical object, represented in code, that can identify patterns in data and provide insights or make decisions about previously unseen data.



A model for our student test scores example

One possibility:

- Build a machine learning model that predicts a student's test score based on their study hours
- Train the model using historical data
- Once it's trained, input a new student's study hours to get a predicted test score for that student



5. Interpret Results

- Derive insights and make policy decisions if needed
- Present the results from your analysis to the stakeholders
- Convince People:
 - Explain the specific conclusion and critical findings, in the most understandable manner

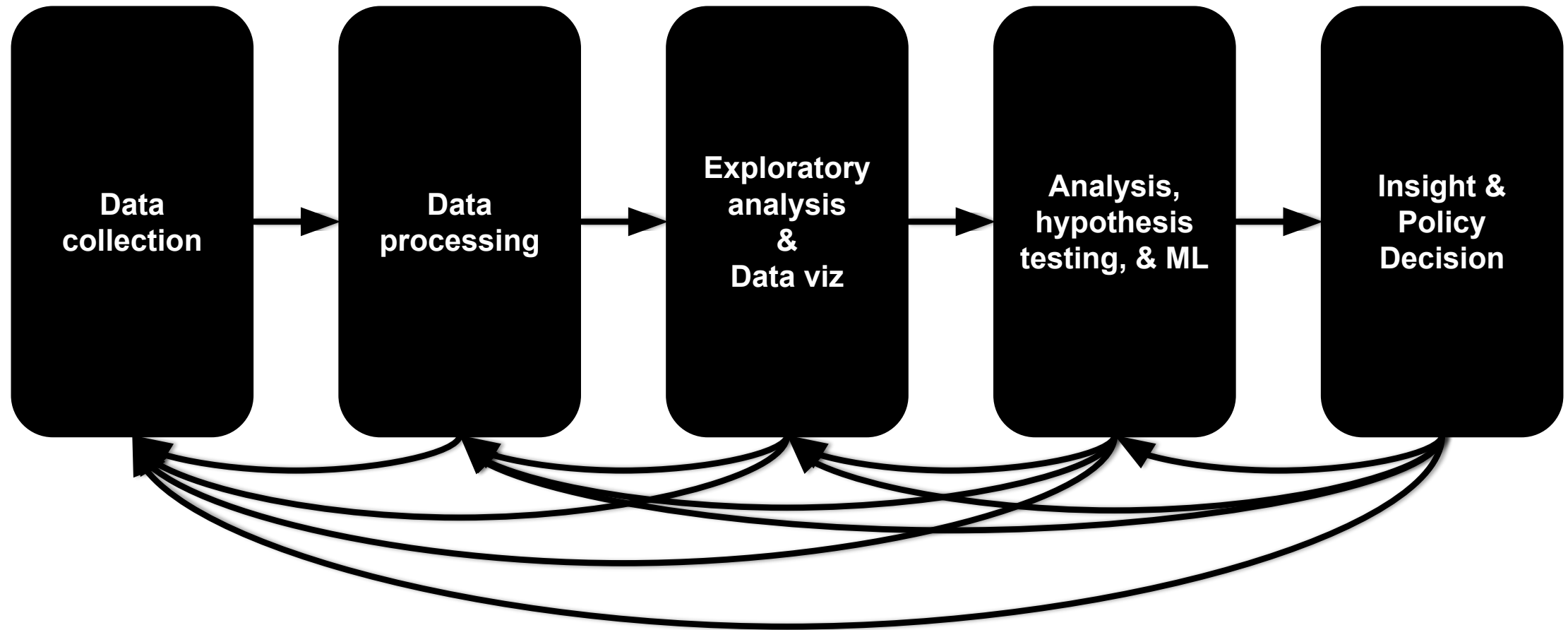


Interpreting our student test scores example

Suppose your analysis confirmed that there's a *strong positive correlation between study hours and test scores*. This insight is crucial for both students and educators to understand the importance of consistent study habits.

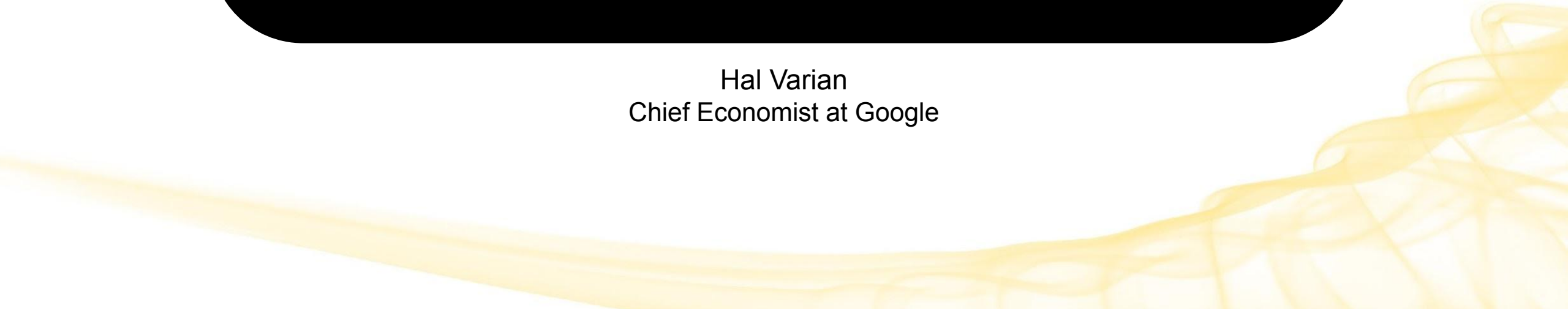
- **Policy Decisions**
 - Your findings could lead to policy decisions that encourage regular study time or improved teaching methods.
- **Convincing people**
 - Often, you must share findings with non-technical groups like marketing or executives. Your goal is clear communication, allowing stakeholders to create actionable plans based on the results.
 - **Be careful!** You also need to make sure they understand what your findings mean and don't come away misunderstanding the results

Important: Data Science is not a linear process!



“The ability to take data – to be able to **understand** it, to **process** it, to **extract value** from it, to **visualize** it, to **communicate** it – that’s going to be a hugely important skill in the next decades ...”

Hal Varian
Chief Economist at Google

A decorative graphic consisting of several thin, flowing, yellow lines that swirl and curve across the bottom right corner of the slide, extending from the bottom edge towards the right edge.

Data vs Big Data

What is Big Data?

Dictionary Definition of Big Data

“Extremely large data sets that may be analyzed computationally to reveal patterns, trends, and associations, especially relating to human behavior and interactions.”

Not particularly helpful. Why?



Naive definition

Big data depends solely on the **data size**

- 1 Gigabyte?
- 1 Terabyte?
- 1 Petabyte?

Good start, but this interpretation misses important aspects:

- Time
 - Diversity
 - Distribution
- 

The 3 Vs of Big Data: Volume, Velocity, Variety

“Big data is high-**volume**, high-**velocity** and/or high-**variety** information assets that demand cost-effective, innovative forms of information processing that enable enhanced insight, decision making, and process automation.”

- Gartner's IT Glossary

The 5 Vs: Volume, Velocity, Variety, Veracity, Value

Volume

- Very high number of data points
 - Scientific data: human genome can be sequenced in a day.
 - Hundreds of millions of tweets per day
 - Exabytes of data created every day

Variety

- Wide range of data types and sources
 - Databases, streams, object storage, spreadsheets, photos, videos, natural text, etc

Velocity

- Speed of data generation, collection and processing
 - Sensors everywhere - can generate high-rate data streams
 - Real-time analytics requires data to be consumed as fast as it is generated

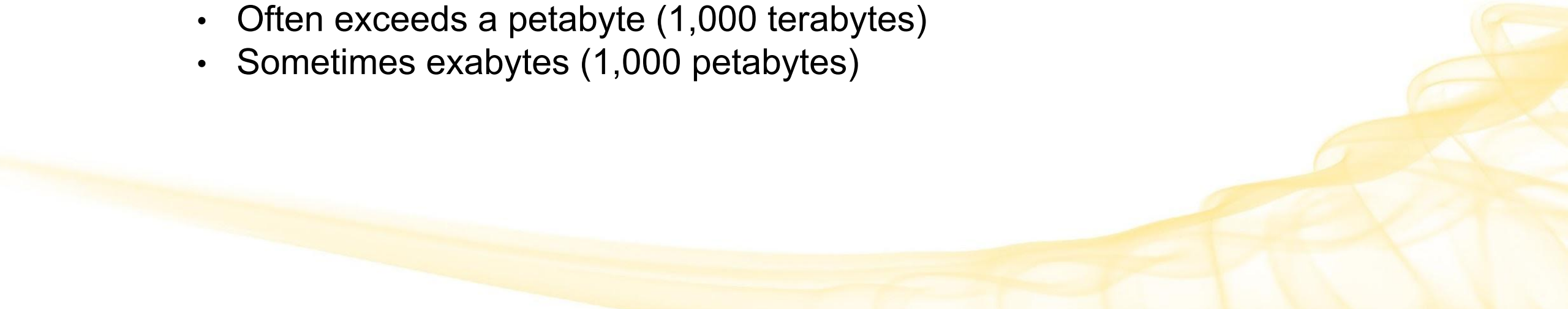
Veracity

- Trustworthiness and quality of the data
 - How do you decide what to trust?
 - How to remove noise?
 - How to fill in missing values?

Value

- The usefulness and importance of the data
 - How can it be used to gain insights and see patterns?

How Big is “Big”?

- No hard definition beyond it being “too big” for traditional computing/processing techniques
 - “Big” is constantly changing as we collect more data and create better techniques to handle it
 - Current “big”
 - At least terabytes (1,000 gigabytes)
 - Often exceeds a petabyte (1,000 terabytes)
 - Sometimes exabytes (1,000 petabytes)
- 

Structured vs unstructured data

Structured data

- Fits into a predefined format and data tables
- Uses discrete data types such as numbers, short text, and dates

Unstructured data

- Doesn't fit neatly into a data table, due to size or nature of format
 - ex. audio, video, large text documents, biometrics
- Can be challenging to analyze and process
- Sometimes, numerical or textual data can be unstructured because modeling it as a table is inefficient
 - Ex. constant stream of numbers from sensors
 - Creating a table with 2 columns, timestamp and sensor value, would be inefficient and impractical

Data Formats

Tabular data

- Things in tables
- Structured data organized into rows and columns
- Often resembles a spreadsheet or database table

Examples:

- Table of grades
- Sales records

Columns stores a specific data type

Row →
Or record

Emp No	Name	Age	Department	Salary
001	Alex S	26	Store	5000
002	Golith K	32	Marketing	5600
003	Rabin R	31	Marketing	5600
004	Jons	26	Security	5100

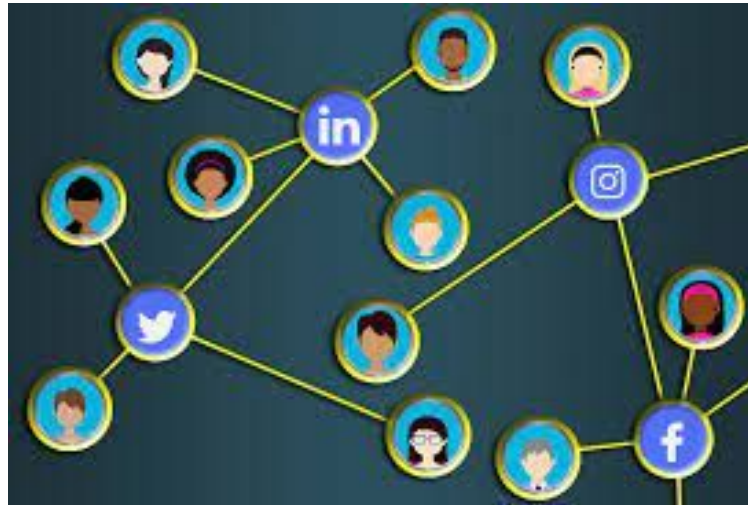
Data Formats

Graph data

- Represents relationships between entities using nodes (vertices) and edges

Examples:

- Social connections
- Network traffic
- Roads



Data Formats

Raw Data

- Data in their collected type format

Examples:

- Image
 - Video
 - Audio
 - Telemetry
- 

Next Monday

Quiz 2!

